Guneev Pannu

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Education

University of Toronto – 3.96 GPA Sept 2024 – May 2028

HBSc Double Major in Computer Science and Statistics, Math Minor

Relevant Courses: Data Structures & Algorithms, Operating Systems, Systems Programming, Databases, Security

Experience

Teaching Assistant Sept 2026 – Dec 2026

University of Toronto *Toronto, Canada*

- Leading weekly tutorials for **40+** students, Python fundamentals (control flow, functions, lists) for CSC108
- Authored weekly lab materials and slide decks in LaTeX to deepen understanding of **500+ students** for CSC108

Product Manager Sept 2026 – Present

UofT Machine Learning Club *Toronto, Canada*

- Leading club website's migration to **Payload CMS**, replacing Google Forms and Git-push content updates with a single self-serve platform for **11 execs** and **100+ applicants** per year
- Interviewed all club VPs across engineering, research, academics, etc, to gather requirements for the website redesign, and combined input into a prioritized roadmap for the year.

Web Developer May 2026 – Aug 2026

Professor - University of Toronto *Toronto, Canada*

- Built a live queuing and logging system using **Next.js**, **React**, **Docker**, and **PostgreSQL** to improve structure in office hours by reducing crowding, and by keeping attendance logs in case of cheating in **500+ student** class
- Worked in a team of 4, meeting weekly with supervising professor to discuss progress and decide next steps
- Wrote GitHub Action to automatically publish updates via self hosted runner to save time during development
- Integrated UofT SAML authentication for **15K+ students**, eliminating all manual account creation

Projects

Temporary RAG | *Python, PostgreSQL, AWS Bedrock*

- Built an ephemeral RAG app for temporary research paper workspaces: users upload PDFs, chat against their corpus with page-level citations, while workspaces auto delete via TTL cleanup on AWS
- Implemented hybrid search using pgvector and BM25 to match both semantic meaning and keywords to more accurately retrieve the relevant chunks related to a user query
- User uploaded files are stored in AWS S3, responses are generated through Haiku from AWS Bedrock
- Documents are chunked on markdown boundaries and indexed with meta data to better maintain semantic structure for better retrieval

AMD Schola Godot Port | *Godot, C++, GDExtension, gRPC*

- Built a C++ RL environment framework and plugin for Godot using GDExtension
- Implemented lifecycle state machines, reset logic, action routing, and auto-reset modes
- Connected Godot environments to Python training tools through a gRPC interface for live episode execution

In Class Live QA Tool | *Next.js, Typescript, PostgreSQL, Redis, Docker*

- Architected a real-time UofT Q&A platform in **Next.js** securing a Fall rollout targeting **500+ students**
- Implemented quality of life features like notifications, and automatic question context with **React Hooks** to make it easier to monitor chat during lecture
- Streamed real time events using **WebSockets** to minimize delay in room events such as question posts, question upvotes, slide changes, and population count in **50+ student** rooms with average latency of **40ms** during load test

Technical Skills

Languages: Typescript, C++, Java, Python
Frameworks: Next.js, React, Node.js, JUnit, Spring Boot, Agile
Developer Tools: Cursor, Codex, Git, Docker, PostgreSQL, Postman, Linux
Libraries: pytest, Prisma, PyTorch, JavaFX